INCH-POUND
MIL-P-24441/33(SH)
23 July 1991

MILITARY SPECIFICATION SHEET

PAINT, EPOXY-POLYAMIDE, DARK GRAY Ro3.6, FORMULA 154, TYPE IV

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-P-24441.

FORMULA. This formula covers dark gray Ro3.6 epoxy-polyamide paint designated Navy Formula 154, type IV, for interior or exterior use. The paint shall consist of the ingredients specified in the quantities specified.

Component A

Ingredients	Pounds
Thixotrope 1/	---
Polyamide 2/	36
Polyamide adduct 3/	309
Magnesium silicate 4/	300
Titanium dioxide 5/	20
Black iron oxide 6/	150
Butyl alcohol 7/	258

Component B

Ingredients	Pounds
Thixotrope 8/	---
Epoxy resin 9/	512
Magnesium silicate 4/	390
Naphtha 10/	208

See footnotes on next page.

AMSC N/A FSC 8010
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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QUANTITATIVE REQUIREMENTS. The paint shall meet the following quantitative requirements and the qualitative requirements of section 3 of the general specification. Components A and B shall be mixed 1:1 by volume for mixed component tests. Tests shall be performed in accordance with the general specification.

Requirements	Component A		Component B		Mixed components	
	Min	Max	Min	Max	Min	Max
Pigment content, percent	41.0	46.0	33.0	38.0	---	---
Volatiles, percent	23.0	28.0	16.0	21.0	---	---
Nonvolatile vehicle, percent	29.0	34.0	44.0	49.0	---	---
Water, percent	---	1.5	---	0.5	---	---
Coarse particles, percent	---	0.3	---	0.3	---	0.5
Consistency, grams	190	260	300	470	180	245
Pounds per gallon (lb/gal)	10.9	11.4	10.8	11.3	10.9	11.4
Set to touch, hours						
(a) at 40°F	---	---	---	---	---	3
(b) at 73°F	---	---	---	---	---	3
Dry-hard, hours						
(a) at 40°F	---	---	---	---	---	24
(b) at 73°F	---	---	---	---	---	8
Fineness of grind, NS	4	---	4	---	4	---
Flash point, °F	99	---	100	---	99	---
Titanium dioxide, percent of pigment	4.0	---	---	---	---	---
Pot life, hours at 73°F	---	---	---	---	6	---
Sag resistance, mils	---	---	---	---	12	---
Color dry film, Rw percent	---	---	---	---	1.74	2.06
Weight per epoxy, vehicle	---	---	175	195	---	---
Contrast ratio, 3 mil dry film	---	---	---	---	0.98	---
Epichlorohydrin content (vehicle), parts per million	---	---	---	2	---	---
Gloss 60 degree specular percent	---	---	---	---	---	30
VOC, ASTM D 3960, lb/gal 11/	---	---	---	---	---	2.8

1/ Thixotrope to be used is the manufacturer's choice. In the development of the component A, 15 pounds (2.1 gallons) of Dislon NS-30 made by King Industries were used. Manufacturer is responsible for choosing a thixotrope that meets all the requirements herein, including shelf life. Thixotrope is a pigment for calculation purposes.

2/ GENAMID 2000, General Mills Chemicals; UNIREZ 2810, Union Camp; AZAMIDE 600, AZ Products; ANCAMINE 507, Pacific Anchor Chemical Corp.; EPOTUF SF7791, Reichhold Chemical.

3/ Xu-HY-283, Ciba-Geiby Corporation.

4/ Mistron 400, "Mistron 500", Cyprus Industries.

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5/ Titanium dioxide conforming to ASTM D 476, type IV. In the development of component A, Dupont Tipure R960 was used.
6/ Black Iron Oxide Bk-5599, Charles Pfizer.
7/ Conforming to ASTM D 304.
8/ Thixotrope to be used is the manufacturer's choice. In the development of the component B, 15 pounds (1.8 gallons) of Dislon 6500 made by King Industries were used. Manufacturer is responsible for choosing a thixotrope that meets all the requirements herein, including shelf life. Thixotrope is a pigment for calculation purposes.
9/ EPON 828, Shell Chemical Co.; ARALDITE 6010, Ciba-Geigy Corp.; DER 331, Dow Chemical Corp.; AZEPOXY 128, AZ Products; EPO-TUF 37-140, Reichhold Chemical; EPI-REZ 510, Celanese.
10/ Conforming to ASTM D 3734, type I. In the development of component B, AMSCO Super High Flash Naphtha was used.
11/ For VOC calculations, component A makes approximately 101 gallons and component B makes approximately 101 gallons.

Use of alternate ingredients in this formula must have prior approval of the Naval Sea Systems Command. Approval will be based on review of data demonstrating equivalent physical, chemical and performance characteristics of paint manufactured with the proposed alternate material. Development of the required data package is the responsibility of the proposing agent. Paint incorporating the proposed alternate ingredient shall conform to all the requirements of this military specification sheet and the general specification.

Preparing activity:
Navy - SH
(Project 8010-N252)

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